JetStock

Inventory & Product Management Application

Capstone Project | Software Engineering Program | April 2026

Products

Full CRUD

Inventory

Live tracking

Suppliers

Contacts

The Problem & Our Solution

❌ The Problem

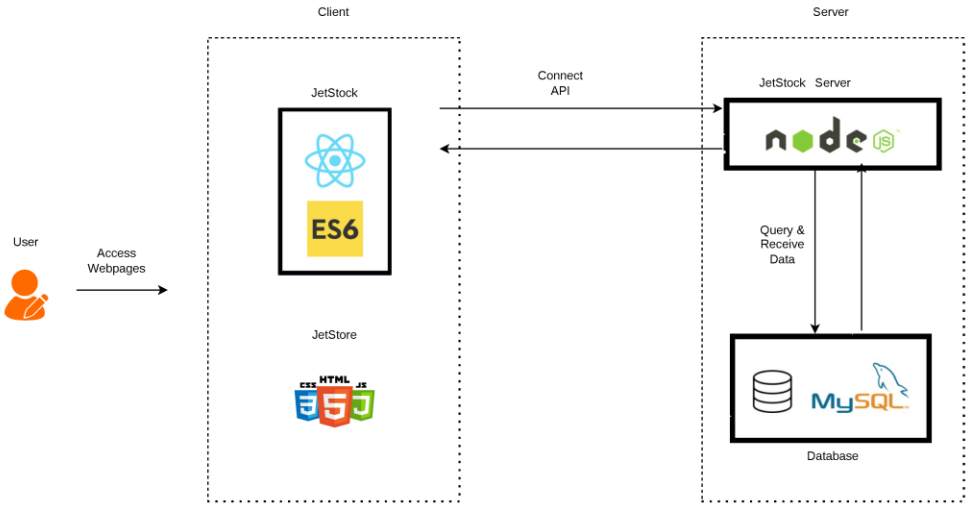
- Manual tracking via notebooks & spreadsheets
- No real-time stock level visibility
- Missing low-stock alerts until out of stock
- No central system — disorganised records
- Existing tools are too complex & expensive
- Time wasted searching for product info

→

✅ JetStock Solution

- Central web app for all inventory data
- Live stock quantity tracking
- Low-stock alerts & reorder level visibility
- Organised by product, category & supplier
- Simple, lightweight — built for small business
- Fast search & filter for any product

System Architecture & Tech Stack



Key Features

Authentication

JWT-based login & registration with role-based access

Product Management

Full CRUD for products with SKU, price, category & supplier

Categories

Organise products into categories for easy filtering

Suppliers

Manage supplier contacts linked to your products

Inventory Control

Real-time quantity tracking with low-stock & out-of-stock alerts

Stock Movements

Log stock IN/OUT with movement type: manual, online, return, damage

Orders

Create and track customer orders with full order item management

Dashboard

Summary view of total products, low-stock count and inventory stats

Database Design (ERD)

Users

id, first_name, last_name, email, role

Products

id, sku, name, price, supplier_id, category_id

Categories

id, name, description, user_id

Stock_movements

id, product_id, movement_type, source_type, qty

Inventory

id, product_id, quantity_on_hand, reorder_level

Suppliers

id, name, contact_person, email, phone

Orders

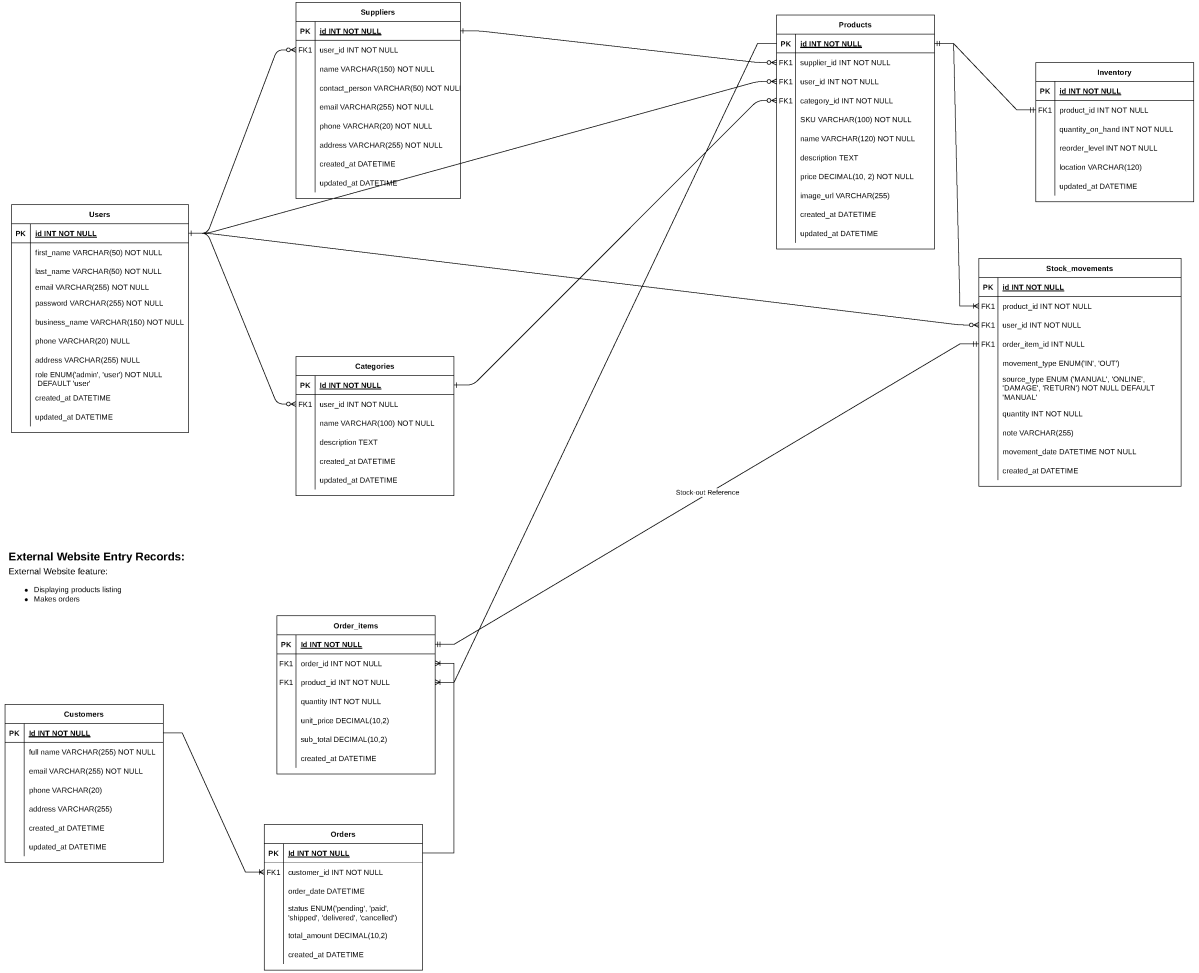
id, customer_id, status, total_amount

Order_items

id, order_id, product_id, quantity, unit_price

Customers

id, full_name, email, phone, address



External Website Entry Records:

External Website feature:

- Displaying products listing
- Makes orders

Project Planning & Timeline

Phase 1 Mar 13–15	Phase 2 Mar 16–24	Phase 3 Mar 24–28	Phase 4 Mar 28–31
Planning & Design	**Backend Dev**	**Frontend Dev**	**Polish & Delivery**
✓ Define project ✓ User stories ✓ Architecture diagram ✓ Lo-fi wireframes	✓ Models & DB setup ✓ CRUD APIs ✓ Auth / JWT ✓ Error handling ✓ Unit tests	✓ Login/Signup ✓ Dashboard ✓ All listing pages ✓ Forms & search ✓ Connect API	✓ UI polish ✓ Frontend tests ✓ Documentation ✓ PPT slides

Space / List

Add Channel | Calendar | Gantt | Doc | Chat | Video Embed | List View | Board | Whiteboard | + View

Today | Day | Auto fit | Export

Name	Due Date	Start Date	Status
List			
Define project	Mar 15 2023	Mar 13 2023	COMPLETE
Write problem statement	Mar 15	Mar 13	COMPLETE
Create user stories	Mar 15	Mar 13	COMPLETE
Create user flow	Mar 15	Mar 13	COMPLETE
Create architecture diagram	Mar 15	Mar 13	COMPLETE
Design lo-fi wireframes	Mar 15	Mar 13	COMPLETE
Create GitHub Repo and Readme	Mar 16	Mar 16	COMPLETE
Setup Backend	Mar 17	Mar 16	COMPLETE
Create Models	Mar 18	Mar 16	COMPLETE
Suppliers CRUD	Mar 21	Mar 19	COMPLETE
Categories CRUD	Mar 21	Mar 19	COMPLETE
Product CRUD	Mar 21	Mar 19	COMPLETE
Stock Movement feature	Mar 23	Mar 22	COMPLETE
Customer Creation CRUD	Mar 23	Mar 22	COMPLETE
Order Creation CRUD	Mar 23	Mar 22	COMPLETE
Authentication/Authorization Feature	Mar 20	Mar 18	COMPLETE
Validation and error handling	Mar 22	Mar 22	COMPLETE
Begin Backend tests	Mar 24	Mar 23	COMPLETE
Setup Frontend	Mar 24	Mar 24	COMPLETE
Login/signup	Mar 24	Mar 24	COMPLETE
Create Dashboard	Mar 25	Mar 24	COMPLETE
Listing pages	Mar 26	Mar 25	COMPLETE
Add/Edit forms	Mar 26	Mar 25	COMPLETE
Connect frontend to backend	Mar 27	Mar 26	COMPLETE
Add search/filter	Mar 27	Mar 26	COMPLETE
Error Handling	Mar 27	Mar 26	COMPLETE
Reports page	Mar 27	Mar 26	COMPLETE
Paginations	Mar 28	Mar 26	COMPLETE
Polish UI	Mar 27	Mar 26	COMPLETE
Frontend tests	Mar 28	Mar 27	COMPLETE
Update Documentation	Mar 30	Mar 28	COMPLETE
PPT slides	Mar 31	Mar 29	COMPLETE

Testing Strategy

Unit Testing (Jest)

- CRUD controller tests
- Stock movement logic
- Auth middleware validation
- Error handling responses

API Testing (Postman)

- All REST endpoints verified
- Valid & invalid request bodies
- Auth headers & protected routes
- HTTP status code verification

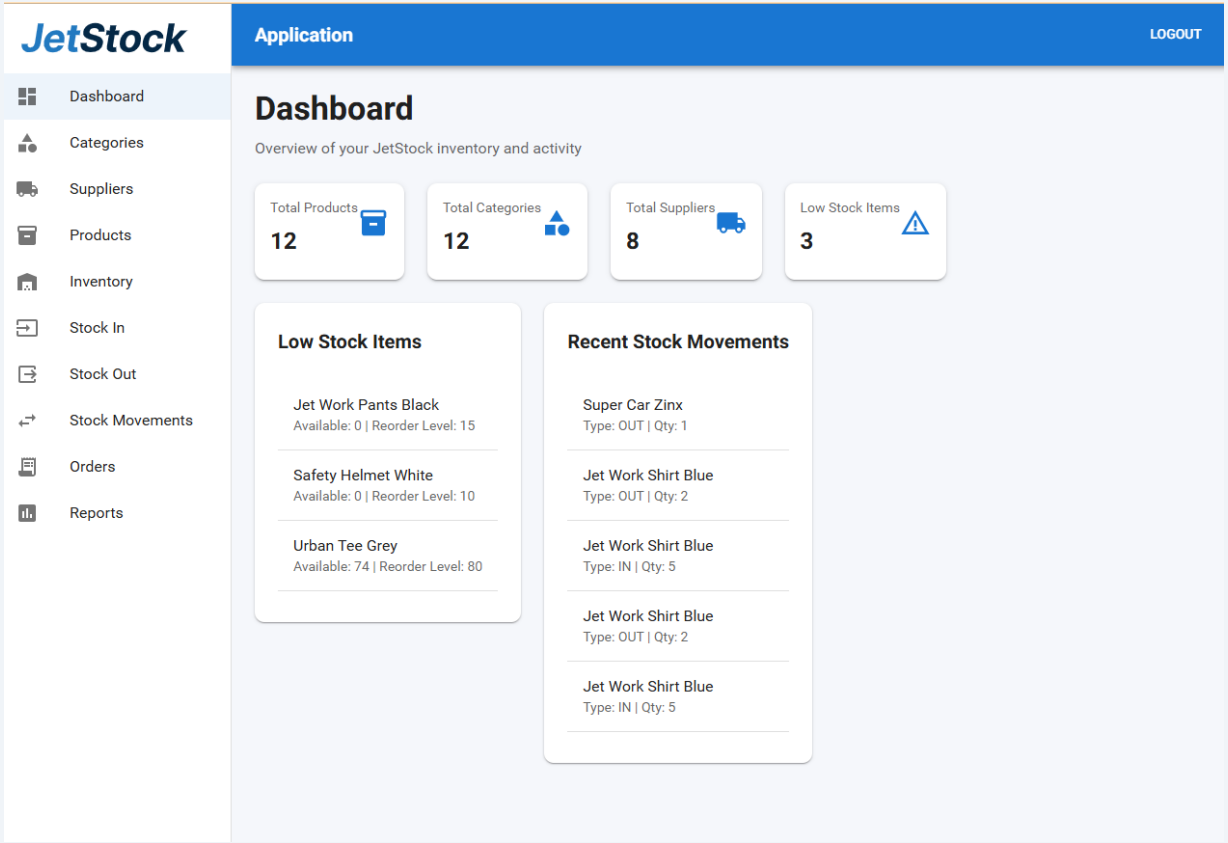
Manual UI Testing

- Login/Registration flow
- All CRUD form validation
- Dashboard real-time data
- Search, filter & pagination

Edge Cases Handled

- Stock out > available qty
- Duplicate SKU detection
- Delete with linked records
- Out-of-stock visual alerts

Application Walkthrough



Project Outcomes

What JetStock delivered

- ✅ Full-stack web app with React + Node.js + MySQL
- ✅ Complete CRUD for Products, Categories, Suppliers & Orders
- ✅ Real-time inventory tracking with low-stock alerts
- ✅ JWT authentication with protected routes
- ✅ External storefront (JetStore) connected via shared API
- ✅ Unit tests (Jest) + API tests (Postman)

https://github.com/mensonsundash/jetstock-capstone